Unit no. ° 1

ARITHMETIC

2-11 September 2026

I Integers, multiples and divisors

Definitions

$\mathbb{N} = \{0; 1; 2; \ldots\}$ is the set of natural numbers. $\mathbb{Z}$ contains the natural numbers and their opposites. For $a, b \in \mathbb{Z}$, saying a is a **multiple of** b means that there exists $k \in \mathbb{Z}$ such that $a = bk$. We say also that b divides a.

$18 = 6 \times 3$: 18 is a multiple of 3 and of 6. Every integer is a multiple of 1 and of itself; 0 is a multiple of every integer. The only multiple of 0 is 0.

Example 1.1

List the positive divisors of 18 and 24. Write three multiples of 7, one of them negative. Is 35 a multiple of 5? of 6? Justify.

Solution:

Divisors of 18: 1 , 2, 3, 6, 9, 18. Divisors of 24: 1 , 2, 3, 4, 6, 8, 12, 24. For example $-7, 0, 14$ are multiples of 7. 35 = 5 $\times$ 7 is a multiple of 5; 35 = 6 $\times$ 5 + 5 is not a multiple of 6.

Sum of multiples

The sum of two multiples of an integer a is a multiple of a.

Write a proof.

Solution:

Let us write $u = ak$ and $v = a\ell$, with $k, \ell \in \mathbb{Z}$. Then $u + v = a(k + \ell)$ and $k + \ell \in \mathbb{Z}$. Hence $u + v$ is a multiple of a. This also applies to $a = 0$.

Fluency · 5 to 10 minutes

Calculate $\frac{2}{3} + \frac{1}{6}$; $3(x + 2) - x$; the image of 2 under $x \mapsto 2x - 3$.

Answers: $\frac{5}{6}$; $2x + 6$; 1.

II Parity and prime numbers

Definitions

An integer n is **even** if it is written $2k$, with $k \in \mathbb{Z}$, and **odd** if it is written $2k + 1$. A **prime number** is a natural number greater than or equal to 2 that has exactly two positive divisors: 1 and itself.

The prime numbers below 100 are: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97.

Property to be proved

The square of an odd integer is odd.

Write a proof.

Solution:

If $n = 2k + 1$, then $n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$. Since $k^2 + 2k \in \mathbb{Z}$, n^2 is odd. The final form justifies the parity; a few numerical trials are not enough to prove the statement.

Test whether an integer is prime

For $n \geq 2$, it suffices to test the prime divisors less than or equal to $\sqrt{n}$. Indeed, if $n = ab$ with $1 < a \leq b$, then $a \leq \sqrt{n}$; a has a prime divisor that also divides n.

Example 1.2

Show that 157 is prime and that 161 is not.

Solution:

Since $12^2 < 157 < 13^2$, we test $2, 3, 5, 7, 11$. None divides 157: for 7, $157 = 7 \times 22 + 3$; for 11, $157 = 11 \times 14 + 3$. So 157 is prime. However, $161 = 7 \times 23$ is composite.

III GCD and irreducible fractions

Definition and convention

For two integers a, b **not both zero** , GCD (a, b) is the largest of their positive divisors in common. We have GCD $(a, b) = \text{GCD}\ (|a|, |b|)$ and GCD $(a, 0) = |a|$ if $a \neq 0$. By convention, we set GCD(0 , 0) = 0; this last case is not covered by the definition as the greatest divisor.

For example, GCD(18 , 24) = 6, and changing the signs of 18 or 24 does not change this result.

Prime factorisation · accepted

Every natural number greater than or equal to 2 is a product of prime numbers. This decomposition is unique up to the order of the factors.

Example 1.3

Write 140, 150, 630 and 120 as products of primes.

Solution:

$140 = 2^2 \times 5 \times 7$; $150 = 2 \times 3 \times 5^2$; $630 = 2 \times 3^2 \times 5 \times 7$; $120 = 2^3 \times 3 \times 5$. We divide successively by prime numbers and check the final product.

Making a fraction irreducible

For $b \neq 0$, divide the numerator a and the denominator b by GCD (a, b). The resulting fraction has a numerator and a denominator that are coprime.

Example 1.4

Make irreducible $\frac{140}{630}$, $\frac{84}{126}$ & $\frac{-120}{150}$.

Solution:

The GCDs are 70, 42 and 30 respectively. Thus $140/630 = 2/9$, $84/126 = 2/3$ and $-120/150 = -4/5$.
For 140 /630, we can also cancel the factors 2 , 5, 7 in the factorisations.

IV Set vocabulary in context

Definitions and notation

$x \in A$ means that x is an element of A ; $A \subseteq B$ means that any element of A belongs to B. $\varnothing$ is the empty set.

- $A \cap B$: elements belonging to A **and** to B ;
- $A \cup B$: elements belonging to A **or** to B (possibly both);
- in a reference set E, $\overline{A} = E \setminus A$ contains the elements of E which are not in A ;

· for a finite set, Card(A) is its number of elements.

Example 1.5

In $E = \{1; 2; 3; 4; 5; 6\}$, set $A = \{2; 4; 6\}$ and $B = \{3; 6\}$. Give $A \cap B$, $A \cup B$, $\overline{A}$ and $\text{Card}(A)$. Are assertions $2 \in A$, $\{2\} \subseteq A$ and $A \subseteq B$ true?

Solution:

$A \cap B = \{6\}$, $A \cup B = \{2; 3; 4; 6\}$, $\overline{A} = \{1; 3; 5\}$, $\text{Card}(A) = 3$. The first two assertions are true. The third is false, for example $2 \in A$ and $2 \notin B$.

Pair and Cartesian product

A pair (a, b) is ordered: in general ($a, b) \neq (b, a)$. The Cartesian product $A \times B$ is the set of the pairs (a, b) with $a \in A$ and $b \in B$. For finite sets, $\text{Card}(A \times B) = \text{Card}(A)\,\text{Card}(B)$.

Example 1.6

Write all the elements of $\{1; 2\} \times \{3; 4; 5\}$.

Solution:

$\{(1, 3); (1, 4); (1, 5); (2, 3); (2, 4); (2, 5)\}$. There are $2 \times 3 = 6$ pairs.

V Algorithms and reuse

Python: integer, boolean and condition

`a % b` is the remainder of the Euclidean division for integers and $b \neq 0$. A test returns a boolean: `True` or `False`. Equality is tested by `==` ; assignment uses `=`.

```
def est_multiple(a, b):
    if b == 0:
        return a == 0
    return a % b == 0
```

Example 1.7

Predict the results of `est_multiple(35, 5)` and `est_multiple(35, 6)`, then of `est_multiple(0, 7)`, `est_multiple(7, 0)` and `est_multiple(0, 0)`. Explain the role of the condition.

Solution:

The results are `True`, `False`, `True`, `False`, `True`. The case $b = 0$ avoids computing a remainder modulo zero, impossible, and applies the definition: only 0 is a multiple of 0.

Example 1.8: Problem

We have 84 exercise books and 126 pencils. We want to make as many identical batches as possible using all the material. How many batches can be made? What does each of them contain?

Solution:

The number of batches divides 84 and 126. Its maximum is therefore GCD $(84, 126) = 42$. Each batch contains $84/42 = 2$ exercise books and $126/42 = 3$ pencils. Checking $42 \times 2 = 84$, $42 \times 3 = 126$ confirms that all the material is used.

Guidance for the teacher

Diagnostic: have students state precisely "divides" and "is a multiple of". Hint: start by testing 2, 3, 6 batches. Extension: prove that $n(n+1)$ is even by distinguishing the two parities of n.